



9498 - Correcting charge migration in the MIRI detectors for compact sources

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The detectors in MIRI can experience significant charge migration. Bright compact sources create strong pixel-to-pixel spatial gradients, especially at the shortest wavelengths where spatial resolution is best. The most intensely illuminated pixels can be effectively de-biased relative to neighboring pixels, and the subsequent difference in electric fields induces photoelectrons to migrate. Point spread functions (PSFs) for bright sources are thereby broader than for faint sources, called the "brighter-fatter effect" (BFE). BFE can create systematic error in spectra of transiting exoplanets, affecting the inference of cloud opacity and molecular abundances. BFE can also affect the spectra of other compact sources such as T-Tauri stars with close-in accretion disks, and the nuclei of some galaxies. In those latter cases where gas emission lines are superposed on a dusty continuum, BFE can affect the ratios of line-to-continuum intensity, potentially altering the inferred physics of the source. The `calwebb-detector1` pipeline can flag the presence of BFE during data reduction, but not correct it. The MIRI instrument team is working on correcting BFE using deconvolution approaches. We propose a complementary method that will be simple, fast, and robust for the spectra of compact sources. Our method uses optimal PSF-weighting in the spatial direction at the group level. This has recently been demonstrated to successfully correct spatial BFE in MIRI/MRS data. We will develop, benchmark, document, and distribute an open source Python code that will be applicable to a wide range of past and future science

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using MIRI, both MRS and LRS.

OBSERVING DESCRIPTION

There is no observing; this an Archival Data Science Software proposal.